

Pokhara University
Faculty of Management Studies

Course Code: CMP 171

Course title: **Fundamentals of Computer Systems**

Nature of the course: Theory + Practical

Year: First, Semester I

Level: Bachelor

Full marks: 100

Pass marks: 45

Credit Hrs: 3

Total periods: 48 hours

Program: BCSIT

1. Course Description

The course aims to provide a comprehensive understanding of computer systems, covering hardware-software interactions, operating systems, data communication, internet services, database management, multimedia, security, privacy, and current computing trends. The course includes study of computer components, CPU, memory, grasp data processing, interactions, software types, and system vs. application, enhance functionality, user experience OS functions, protection, installation multimedia, graphics, sound, image and demonstrate practical knowledge.

General Objectives

- To explore the historical evolution and significance of computers in modern society.
 - To gain insight into computer hardware components, including CPUs, memory, storage, and I/O devices.
 - To understand different types of software, such as operating systems and applications software.
 - To examine the role of operating systems in managing hardware and user interactions.
 - To learn about network protocols, transmission media, and architectures enabling seamless digital connectivity.
 - To explore the structure, services, and transformative impact of the Internet on global communication and commerce.
 - To acquire skills in designing, manipulating, and securing data using database management systems.
 - To explore the world of multimedia, learn critical aspects of computer security and privacy, and gain insights into emerging technologies and their transformative impact on various industries.
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3.Method of Instructions

General Instructional Technique: Lecture, Discussion, Readings, Question Answer

Specific Instructional Technique: Practical works, Project Based Learning, Self-Directed Learning, Industry Insights, Field Visit and Case Study

4. Content in Detail with Specific Objectives

Specific Objectives	Contents
- Explore computer's characteristics, history, and generational progression, highlighting their roles in modern society. - Explain data representation and various computer applications across fields like business, communication, and research.	Unit 1: Introduction to Computer [6 Hrs.] 1.1 Introduction 1.2 Characteristics of Computer 1.3 History of Computing Devices 1.4 Generations of Computer 1.5 Classification of Computer 1.6 Data and Program Representation in Computer 1.7 Applications of Computers
Specific Objectives	Contents
- Define digital computer components - ALU, CU, Memory, I/O. Illustrate functional block diagram, demonstrating their coordinated operation. - Explore into the significance of each component, their roles in data processing, and their collective contribution to overall computing.	Unit 2: Computer Hardware [4 Hrs.] 2.1 Introduction 2.2 Functional Block Diagram of Digital Computer System 2.2.1 ALU, CU, Memory Unit, Input and Output Units
Specific Objectives	Contents
- Differentiate between system (development, management) and application software (package, tailored), illustrating their vital roles. - Highlight software's diverse applications, from system control to tailored solutions addressing specific business needs.	Unit 3: Computer Software [2 Hrs.] 3.1 Introduction 3.2 Types of Software, (System Software (System Development, System Management), Application Software (Package, Tailored))

Specific Objectives	Contents
▪ Clarify OS objectives and types, including interactive, batch, real-time. Discuss their critical role in managing resources. ▪ Detail OS functions - process, memory, file, device management, security, and user interface - driving efficient computing.	Unit 4: Operating System [6 Hrs.] 4.4 Introduction 4.5 Objective of Operating System 4.6 Types of OS 4.7 Functions of OS 4.8 Process Management 4.9 Memory Management 4.10 File Management 4.11 Device Management 4.12 Protection and Security) 4.13 User Interface
Specific Objectives	Contents
▪ Elaborate on communication media, modes (simplex, half-duplex, full-duplex), illustrating how networks facilitate information exchange. ▪ Discuss various network types (LAN, WAN), topologies (star, mesh), and protocols (IP, TCP), crucial for modern interconnected systems.	Unit 5: Data Communication and Computer Network [5 Hrs.] 5.1 Introduction 5.2 Communication Media 5.3 Communication Mode 5.4 Computer Network 5.5 Advantage and disadvantage of Computer Network 5.6 Types of Networks 5.7 Network Topology 5.8 Communication Protocols (IP, TCP) 5.9 Networking Hardware
Specific Objectives	Contents
▪ Trace Internet's evolution, mechanics of data exchange via TCP/IP, and explain IP addressing and DNS systems. ▪ Examine the structure of client-server architecture, key protocols (HTTP, SMTP, POP), and e-commerce, fostering global connectivity.	Unit 6: Internet and Internet Services [4 Hrs.] 6.1. Introduction 6.2. History of Internet 6.3. Working Mechanism of Internet

	6.4. IP Address and Domain Name System (DNS) 6.5. Client-Server Architecture 6.6. Internet Protocols (HTTP, SMTP, POP, FTP, TELNET) 6.7. Static and Dynamic Web Pages 6.8. Search Engines 6.9. E-Commerce and M-Commerce 6.10. E-Governance
Specific Objectives	Contents
- Introduce databases and their attributes, highlight relational database management systems (RDBMS), and discuss data models. - Contrast databases with file systems, and reveal DBMS applications, underlining their pivotal role in efficient data management.	Unit 7: Database Management System [5 Hrs.] 7.1.Introduction to Database 7.2.Data Concepts and Characteristics 7.3.Database Vs. File System 7.4.Data Models 7.5.Database Management System 7.6.RDBMS 7.7.Database Applications
Specific Objectives	Contents
- Get insights into multimedia elements - graphics, sound, image formats, and how they compose immersive experiences. - Explore multimedia's applications, from web-based presentations to the emerging realms of augmented and virtual reality.	Unit 8: Multimedia [4 Hrs.] 8.1. Introduction 8.2. Elements of a multimedia system (Graphics, Sound, Image File Format) 8.3. Web Based Multimedia 8.4. Applications of Multimedia 8.5. Concept of Augmented and Virtual Reality.
Specific Objectives	Contents
Address computer security concerns, access control mechanisms, unauthorized access prevention, and ethical considerations.	Unit 9: Computer Security and Privacy [6 Hrs.] 9.1. Computer Security 9.2. Access Control 9.3. Protecting Against Unauthorized Access

▪ Explain countermeasures against software piracy, malware, and viruses, and delve into network security, encryption, and cyber law.	9.4. Software Piracy 9.5. Computer Viruses (Spyware, Malware, Ransom ware) 9.6. Anti-Virus 9.7. Ethical Issues 9.8. Cyber Law 9.9. Network Security 9.10. Hardware and Software Firewall 9.11. Data and Message Security 9.12. Encryption and Decryption
Specific Objectives	Contents
▪ Unpack contemporary trends: big data's impact, AI's applications, machine learning's growth, and the potential of cloud computing. ▪ Introduce blockchain's decentralized paradigm, IoT's connected devices, and how GIS and BI empower data-driven decisions.	Unit 10: Current Trends in Computing [6 Hrs.] 10.1 Data Warehousing and Data Mining 10.2 Big Data 10.3 Data Science 10.4 Artificial Intelligence 10.5 Machine Learning 10.6 Cloud Computing 10.7 Block Chain Technology 10.8 Digital Marketing 10.9 Internet of Things 10.10 Geographical Information System 10.11 Business Intelligence

5. Laboratory Work

This unit encompasses hands-on tasks, from computer assembly to researching technology trends. Skills gained include word processing, spreadsheet manipulation, dynamic presentations, email management, operating system handling, file organization, and troubleshooting. Students develop practical proficiency and strategic insights for computer technology applications and future career planning.

1. Hands-on lab tasks include building and disassembling a computer.
2. Word Processing
Word Processing Basics; Opening and Closing of documents; Text creation and Manipulation; Formatting of text; Table handling; Spell check, language setting and thesaurus; Page Setup and Layout, references (Table of Content, Citation, Captions), Track Change.
3. Spread Sheet
Basics of Spreadsheet; Manipulation of cells; Formulas and Functions (Math, Text, Statistical, LookUp), Pivot Table, Descriptive Analysis
4. Presentation Tool
Hands-on lab tasks include creating and editing presentations, using templates and styles, and incorporating media and animation.
5. Hands-on lab tasks include setting up and configuring email accounts, sending and receiving messages, and managing contacts.
6. Hands-on lab tasks include installing and configuring an operating system, managing users and permissions, and troubleshooting common issues.
7. Hands-on lab tasks include creating and organizing files and folders, using command line tools to navigate and manage files, and backing up and restoring data.
8. Hands-on lab tasks include diagnosing and resolving common computer problems, performing regular maintenance tasks, and creating and restoring backups and remote login.
9. Hands-on lab tasks include research and analysis of current and emerging trends in computer technology and the development of a career plan.

Note:

1. Motivate students to create small project work integrating all of the above concepts.
 2. Each of the above lab sessions should cover more than 4 hours of practical work.
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6. Evaluation system and Student's Responsibility

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Internal Assessment	60%	
		Practical		20
		Attendance & Class Participation	10%	
		Lab Report/Project Report	20%	
		Practical Exam/Project Work	40%	
		Viva	30%	
Total External	50	Total Internal		50
Full Marks: 50 + 50 = 100				

7. Student's Requirements

Each student must secure at least 45% marks separately in both internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear in the Semester-End Examinations. Students are advised to attend all the classes, formal exams, tests, etc., and complete all the assignments within the specified time period. ***Students are required to complete all the requirements defined for the completion of the course.***

8. Prescribed Books and References

Prescribed Text Books:

1. Thareja, R. (2019). *Fundamentals of computers*. Oxford University Press, USA.
2. Goel, A. (2010). *Computer Fundamentals*. Pearson Education India.

Reference Books:

1. Stallings, W., & Brown, L. (2018). *Computer security: Principles and Practice*. Pearson.
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2. Norton. (2010). *Introduction To Computers (Sie)*. Tata McGraw-Hill Education.
3. Sinha, P. K., & Sinha, P. (2004). *Computer Fundamentals*. BPB publications.
4. Han, J., & Kamber, M. (2012). *Data mining: Concepts and Techniques*. Morgan Kaufmann.
5. Sosinsky, B. (2011). *Cloud Computing Bible*. John Wiley & Sons.